Asset Pricing Theory: S05

Asset Pricing with Frictions

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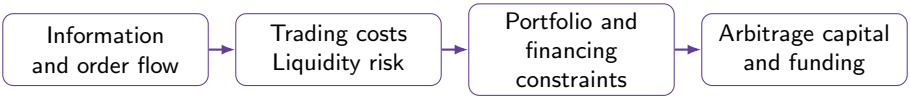
Structure of the course

- Email: amedeo.andriollo@dauphine.psl.eu
- Grading follows a 30-70 rule: 70% final exam, 30% mid-term exam.
- Office hours: feel "free" to (DO) come: B612.
- Slides/materials are updated regularly. Report a typo/mistake: corrections help the class.
- This session:
  - Frictions:information, price discovery, trading costs, liquidity risk, portfolio constraints, and limits to arbitrage.
  - Main references: Glosten-Milgrom (1985) JFE; Kyle (1985) ECTA; Acharya-Pedersen (2005) JFE; Frazzini-Pedersen (2014) JFE; Shleifer–Vishny (1997) JF.

▷ Information and Asset Pricing.

Today's Route: When Trade Is Imperfect

- **Sessions 1–2:** NA and household optimality. **Session 3:** Intertemporal optimality. **Session 4:** Firms' investment decision.
- **This session:** *Who can trade claims, with what information, at what cost, and with what financing?*



- (Frictionless) Benchmark: Investors can adjust positions in either direction at a single price, w/out trading costs or binding financing constraints.

↪ **With frictions:** Trading may reveal information, execution may be costly, desired positions may be infeasible, and arbitrage may be limited.

The Frictionless Benchmark

- **Sessions 1–2:** An investor can buy/sell an additional share at the same market price $P_{i,t}$. The share delivers tomorrow $X_{i,t+1} = P_{i,t+1} + D_{i,t+1}$, so: $P_{i,t} = \mathbb{E}_t \left[M_{t+1} X_{i,t+1} \right], \quad M_{t+1} = \beta u_{C,t+1} / u_{C,t}$.
- **Session 4:** The firm faces the same intertemporal trade-off when choosing physical investment:

$$q_t = \mathbb{E}_t \left[M_{t+1} \left(F_{K,t+1} - G_{K,t+1} + (1-\delta) q_{t+1} \right) \right].$$

↪ The investor/firm can adjust the position in either direction.

- Financial trades occur at a **single price, w/out transaction costs or portfolio constraints**.
- Replicating a payoff additionally requires that the necessary portfolio can be constructed and financed.

- With frictions we may add different dimensions:
  - **Information:** Traders may have different information; order flow reveals information and affects quotes.
  - **Trading costs:** Buying/selling can generate different executable cash flows.
  - **Portfolio constraints:** Some marginal trades are infeasible (moving from equalities to inequalities).
  - **Scarce funding:** Financing a position carries an additional cost.

↪ A friction can change the price *w/out changing its underlying payoff/fundamental*.

⊵ What changes is the ability/cost/info available to trade.

Glosten-Milgrom (1/4): Learning from Trade

- *Can trading change asset prices when the fundamental value has not changed?*
  Shock to fundamental changes the payoffs; informative trade changes what mkt participants **know** about them.
- Consider an asset with uncertain terminal value: $V \in \{V_L, V_H\}, \quad V_H > V_L, \quad q = \Pr(V = V_H) \in (0,1)$.
  Nature draws first V and then a trader's type.

  A competitive risk-neutral dealer posts an ask and a bid *before observing the order*.
  - **Informed** trader (probability $\alpha \in [0,1)$): observes V and buys one share if $V = V_H$, sells one if $V = V_L$.
  - **Uninformed** trader (probability $1-\alpha$): buys/sells one share with prob. $1/2$ each, independently of V.
  - The **dealer** observes the **direction of the order**, neither the trader's type nor V.
  - **Ask**: lowest price a seller is willing to trade VS. **Bid**: the highest price a buyer is willing to pay.
- Denote B buy and S sell:

	$V = V_H$	$V = V_L$
$\Pr(B \mid V)$	$\alpha + \frac{1-\alpha}{2} = \frac{1+\alpha}{2}$	$\frac{1-\alpha}{2}$
$\Pr(S \mid V)$	$\frac{1-\alpha}{2}$	$\alpha + \frac{1-\alpha}{2} = \frac{1+\alpha}{2}$

↪ **Economic mechanism:** When $\alpha > 0$, a buy is more likely if $V = V_H$, while a sell is more likely if $V = V_L$. Uninformed orders ($\alpha \to 0$) prevent the dealer from inferring V perfectly.

▷ **Adverse selection:** The dealer faces different expected asset values conditional on buying versus selling.

Glosten-Milgrom (2/4): Zero Dealer Profit

- Let $\Delta V = V_H - V_L$ then: (Bayes')

$$q_B \equiv \Pr(V_H \mid B) = \frac{\Pr(B \mid V_H)q}{\Pr(B)} = \frac{(1+\alpha)q}{q(1+\alpha)+(1-q)(1-\alpha)},$$

$$q_S \equiv \Pr(V_H \mid S) = \frac{\Pr(S \mid V_H)q}{\Pr(S)} = \frac{(1-\alpha)q}{q(1-\alpha)+(1-q)(1+\alpha)}.$$

  ↪ For $0 < \alpha < 1$, we have $q_B > q > q_S$: buying raises the posterior probability of a high fundamental value; selling lowers it.

- (Asm.:) the dealer is risk-neutral and earns zero expected profit *conditional on each executed trade*:

$$\underbrace{\mathbb{E}[Ask - V \mid B] = 0}_{\text{dealer sells at the ask}} \implies \boxed{Ask = \mathbb{E}[V \mid B] = V_L + q_B \Delta V},$$

$$\underbrace{\mathbb{E}[V - Bid \mid S] = 0}_{\text{dealer buys at the bid}} \implies \boxed{Bid = \mathbb{E}[V \mid S] = V_L + q_S \Delta V}.$$

- **Why is there a spread?** Since $q_S < q < q_B$: $Bid < \mathbb{E}[V] = V_L + q\Delta V < Ask \implies Ask - Bid = (q_B - q_S)\Delta V > 0$.
  ↪ A buy (sell) signals that V is more likely to be high (sell). What does the dealer need to do?
  To charge a higher ask (lower bid) to avoid systematically selling undervalued (buying overvalued) shares.

↪ **Adverse selection:** *Informed traders profit* by buying when $V = V_H$ and selling when $V = V_L$.
The dealer offsets *losses through gains on uninformed* orders, breaking even on average.

▷ A positive bid–ask spread can therefore arise purely from asymmetric information.

Glosten-Milgrom (3/4): The Spread

- **Symmetric prior:** Suppose $q = \Pr(V_H) = 1/2$, so $\Pr(B) = \Pr(S) = 1/2$. By Bayes' rule:

$$q_B = \frac{1+\alpha}{2}, \qquad q_S = \frac{1-\alpha}{2}.$$

Denoting $\bar{V} = (V_H + V_L)/2$, the competitive quotes become:

$$Ask = \bar{V} + \frac{\alpha \Delta V}{2}, \qquad Bid = \bar{V} - \frac{\alpha \Delta V}{2}.$$

Thus, the **bid–ask spread** is:

$$\boxed{Ask - Bid = \alpha \Delta V = 2\alpha\sigma_V}, \qquad \sigma_V = \Delta V/2.$$

↪ $\alpha \uparrow$: More informed trading makes each order more informative, widening the spread.
↪ $\Delta V \uparrow$: A larger fundamental value gap increases the dealer's potential loss to informed traders.

! **Uncertainty alone is not sufficient:** If $\alpha = 0$, then $q_B = q_S = q$ and $Ask = Bid = \mathbb{E}[V]$, even when $\sigma_V > 0$. Without informed traders, orders reveal no information about V.

≥ The spread depends on both **fundamental uncertainty** (ΔV) and **asymmetric information** (α).

Glosten-Milgrom (4/4): Beyond Symmetry

- Let $q = \Pr(V_H) \in (0,1)$. By the law of total probability:

$$\Pr(B) = q\frac{1+\alpha}{2} + (1-q)\frac{1-\alpha}{2} = \frac{1+\alpha(2q-1)}{2}, \qquad \Pr(S) = \frac{1-\alpha(2q-1)}{2}.$$

  ↪ For $\alpha > 0$ and $q \neq 1/2$, buys and sells are no longer equally likely.

- **Bayesian posteriors:** Define $d = \alpha(2q-1)$. Using the posteriors derived previously:

$$q_B - q_S = q\left(\frac{1+\alpha}{1+d} - \frac{1-\alpha}{1-d}\right) = \frac{q[(1+\alpha)(1-d) - (1-\alpha)(1+d)]}{1-d^2}$$
$$= \frac{2q(\alpha - d)}{1-d^2} = \frac{4\alpha q(1-q)}{1-\alpha^2(2q-1)^2}.$$

- **The general bid–ask spread:** Multiplying the difference in posteriors by ΔV:

$$\boxed{Ask - Bid = \Delta V(q_B - q_S) = \Delta V \frac{4\alpha q(1-q)}{1-\alpha^2(2q-1)^2}.}$$

  ↪ At $q = 1/2$, we recover $Ask - Bid = \alpha\Delta V$.
  ↪ For fixed $0 < \alpha < 1$, the spread is largest at $q = 1/2$ and approaches zero as $q \to 0$ or $q \to 1$.

In-Class Exercise 1: Learning from Trades

- We are in the Glosten-Milgrom economy with: $V_H = 120$, $V_L = 80$, $q = \Pr(V_H) = \frac{3}{5}$, $\alpha = \frac{1}{2}$.
- **Questions (20'):**
  1. **Order probabilities and beliefs:** Construct the likelihood table $\Pr(B \mid V)$ and $\Pr(S \mid V)$. Compute $\Pr(B)$ and $\Pr(S)$, then use Bayes' rule to derive $q_B = \Pr(V_H \mid B)$ and $q_S = \Pr(V_H \mid S)$.
  2. **Competitive quotes:** Using the dealer's conditional zero-profit conditions, compute the ask, bid, and bid–ask spread. Compare both quotes with $\mathbb{E}[V]$. Why do they differ?
  3. **Learning from sequential trades:** Suppose the first customer buys. The dealer updates beliefs, and a new trader arrives without any additional public news.
     - Compute the **new bid** at which the dealer accepts a sell.
     - After observing a buy followed by a sell, what is the posterior probability of V_H?
  4. **More informed trading:** Back to the original prior $q = 3/5$ and increase α to $3/4$. Recompute the quotes and spread. Explain what changes and whether the distribution of fundamental values has changed.

Exercise 1 Solution (1/2): Posteriors and Quotes

- With $\alpha = 1/2$, an informed trader buys only when $V = V_H$, while an uninformed trader buys with probability $1/2$:

	V_H	V_L
$\Pr(B \mid V)$	$3/4$	$1/4$
$\Pr(S \mid V)$	$1/4$	$3/4$

  By the law of total probability: $\Pr(B) = \frac{3}{5}\frac{3}{4} + \frac{2}{5}\frac{1}{4} = \boxed{11/20}$, $\Pr(S) = \frac{3}{5}\frac{1}{4} + \frac{2}{5}\frac{3}{4} = \boxed{9/20}$.

- **Updated beliefs:** Using Bayes' rule, conditional on the observed order:

$$q_B = \Pr(V_H \mid B) = \frac{\Pr(B \mid V_H)\Pr(V_H)}{\Pr(B)} = \frac{9/20}{11/20} = \boxed{\frac{9}{11}}, \quad q_S = \Pr(V_H \mid S) = \boxed{\frac{1}{3}}.$$

  ↪ A buy raises the probability of V_H from $3/5$ to $9/11$; a sell lowers it to $1/3$.

- **Competitive quotes:** Zero expected profit conditional on each trade direction implies:

$$Ask = \mathbb{E}[V \mid B] = 80 + 40\frac{9}{11} = \boxed{112.727}, \quad Bid = \mathbb{E}[V \mid S] = 80 + 40\frac{1}{3} = \boxed{93.333}.$$

  Thus:

$$Ask - Bid = \frac{640}{33} = \boxed{19.394}, \qquad \mathbb{E}[V] = 80 + 40\frac{3}{5} = 104.$$

↪ Why do quotes differ? A buy (sell) signals a higher (lower) fundamental value. Quoting 104 on both sides would expose the dealer to expected losses conditional on each order direction.

Exercise 1 Solution (2/2): Learning and the Spread

- **Sequential learning:** After the first buy, the dealer updates the prior to $\breve{q} = 9/11$. With no new public information, a subsequent sell implies: $q_{BS} = \Pr(V_H \mid B, S) = (\frac{9}{11}\frac{1}{4})/(\frac{9}{11}\frac{1}{4} + \frac{2}{11}\frac{3}{4}) = \boxed{3/5}$ so that: $Bid^{\text{seq}} = 80 + 40q_{BS} = \boxed{104}$.
  $\hookrightarrow$ The buy initially raises the probability of V_H, but the subsequent sell reverses that update.
  Opposite orders cancel their likelihood ratios, restoring the original prior $q = 3/5$.
- **More informed traders:** $q = 3/5$ and α goes from $1/2$ to $3/4$, so: $\Pr(B \mid V_H) = 7/8$ and $\Pr(B \mid V_L) = 1/8$. Hence:

$$\Pr(B) = 23/40, \qquad \Pr(S) = 17/40, \quad q_B = \frac{(3/5)(7/8)}{23/40} = \frac{21}{23}, \quad q_S = \frac{(3/5)(1/8)}{17/40} = \frac{3}{17}.$$

  The competitive quotes and spread become:

$$Ask = 80 + 40\frac{21}{23} = \boxed{116.522}, \quad Bid = 80 + 40\frac{3}{17} = \boxed{87.059}, \qquad Ask - Bid = \boxed{29.463}.$$

  $\hookrightarrow$ The spread increases from 19.394 to 29.463: orders are more informative, while the fundamental values and their prior probabilities remain unchanged.

! α is the probability of an *informed trader*, not the probability of a buy. Dealer break-even still requires:
$\mathbb{E}[Ask - V \mid B] = 0, \quad \mathbb{E}[V - Bid \mid S] = 0$, which also imply: $\boxed{\Pr(B)Ask + \Pr(S)Bid = \mathbb{E}[V].}$

$\trianglerighteq$ **What to take home:** prices respond to information revealed by orders, not necessarily to changes in fundamental value.

Kyle (1/4): The Insider's Trading Decision

- *What if an informed trader can choose how much to trade, knowing that larger orders affect prices?*
- Consider a single auction with: $V \sim N(\mu, \sigma_V^2), \quad u \sim N(0, \sigma_u^2), \quad V \perp u.$
  - **Insider:** Observes fundamental V and chooses order x (+ive for buy, -ive for sell), w/out observing noise demand u.
  - **Dealer:** Observes only total order flow $y = x + u$ and set the competitive execution price $P(y) = \mathbb{E}[V \mid y]$.
- **Price impact:** Conjecture a linear pricing rule: $P(y) = \mu + \lambda y, \qquad \lambda > 0.$
  $\hookrightarrow$ A larger buy (sell) raises (lowers) the execution price. λ measures the price impact of an additional share.
- **Insider's problem:** Taking λ as given, the risk-neutral insider maximizes expected trading profits:

$$\max_x \ \mathbb{E}[(V - P(x+u))x \mid V] = \max_x \ \mathbb{E}[(V - \mu - \lambda(x+u))x \mid V] = \max_x \left\{ \underbrace{(V-\mu)x}_{\text{information gain}} - \underbrace{\lambda x^2}_{\text{price-impact cost}} \right\},$$

- Optimal trading corresponds to the FOC:

$$V - \mu - 2\lambda x = 0, \quad \Longrightarrow \quad \boxed{x = (V-\mu)/2\lambda}, \qquad \text{with} \quad \partial^2 \mathbb{E}[\pi \mid V]/\partial x^2 = -2\lambda < 0$$

$\hookrightarrow$ The insider trades more aggressively when $|V - \mu|$ is larger, but less aggressively when λ is higher.
An additional share worsens the execution price on the *entire order*.

$\trianglerighteq$ The insider's FOC determines optimal trading given λ (the pricing rule).

Kyle (2/4): What the Dealer Infers

- We established that the insider optimally trades at $x = \beta(V - \mu)$, $\beta = 1/(2\lambda)$.
  How do dealers determine λ from the information contained in total order flow?
- **Observed order flow y:**

$$y = \underbrace{\beta(V-\mu)}_{\text{informed demand}} + \underbrace{u}_{\text{noise demand}}.$$

  - A positive y suggests a high V, but noise trading prevents dealers from perfectly identifying the fundamental value.
  - The informativeness of y depends on the variability of informed demand relative to noise demand.
- **Bayesian inference:** Since V and u are independent Gaussian variables, (V, y) is jointly normal:

$$\mathbb{E}[y] = 0, \quad \mathrm{Cov}(V, y) = \mathrm{Cov}(V, \beta(V-\mu)+u) = \beta\sigma_V^2, \quad \mathrm{Var}(y) = \beta^2\sigma_V^2 + \sigma_u^2.$$

- The conditional expectation of jointly normal variables is linear:

$$P(y) = \mathbb{E}[V \mid y] = \mu + \frac{\mathrm{Cov}(V,y)}{\mathrm{Var}(y)}(y - \mathbb{E}[y]) = \mu + \underbrace{\frac{\beta\sigma_V^2}{\beta^2\sigma_V^2 + \sigma_u^2}}_{\lambda} y.$$

  Hence, the dealer's price-impact coefficient must satisfy: $\lambda = (\beta\sigma_V^2)/(\beta^2\sigma_V^2 + \sigma_u^2)$.

↪ For a given insider strategy β, more trading noise ($\sigma_u \uparrow$) makes each order less informative, reducing the dealer's response λ.

▷ **Competitive pricing:** $P(y) = \mathbb{E}[V \mid y]$ ensures zero expected dealer profit *conditional on observed order flow*, stronger than just $\mathbb{E}[P(y)] = \mu$.

Kyle (3/4): Equilibrium and Market Depth

- We obtained the two sides of the coin:

$$\underbrace{\beta = 1/(2\lambda)}_{\text{insider's optimal trading}} \qquad \underbrace{\lambda = (\beta\sigma_V^2)/(\beta^2\sigma_V^2 + \sigma_u^2)}_{\text{dealer's Bayesian pricing}}.$$

Which trading intensity and price impact make these decisions mutually consistent?

- **Equilibrium:** Substituting the dealer's pricing coefficient into the insider's FOC:

$$\beta\lambda = \frac{1}{2} \quad \Longrightarrow \quad \frac{\beta^2\sigma_V^2}{\beta^2\sigma_V^2 + \sigma_u^2} = \frac{1}{2} \quad \Longrightarrow \quad \boxed{\beta = \frac{\sigma_u}{\sigma_V}, \qquad \lambda = \frac{\sigma_V}{2\sigma_u}},$$

- $\lambda = \partial P/\partial y$ measures price impact, while $1/\lambda = (2\sigma_u)/\sigma_V$ is the amount of order flow required to move the asset's price by one unit, i.e. market depth (deep/thin mkt has small/large λ).
  - ↪ $\sigma_u \uparrow \Longrightarrow \lambda \downarrow$: More noise trading allows larger orders to be absorbed with less price impact.
  - ! But $\beta \uparrow$: The insider exploits this camouflage by trading more aggressively (see next slide).
- **Equilibrium order flow:** Since $x = \beta(V - \mu)$ and $y = x + u$, with $V \perp u$:

$$\mathrm{Var}(x) = \beta^2\sigma_V^2 = (\sigma_u/\sigma_V)^2\sigma_V^2 = \boxed{\sigma_u^2}, \qquad \mathrm{Var}(y) = \mathrm{Var}(x) + \mathrm{Var}(u) = \boxed{2\sigma_u^2}.$$

↪ In equilibrium, informed and noise orders have equal variances. More noise trading induces a proportional increase in informed trading.

▷ **Strategic equilibrium:** Greater noise trading increases market depth, but the insider's response prevents orders from becoming proportionally less informative.

Kyle (4/4): Price Informativeness

- Recall the equilibrium: $\beta = \frac{\sigma_u}{\sigma_V}$, $\lambda = \frac{\sigma_V}{2\sigma_u}$, $\mathrm{Var}(y) = 2\sigma_u^2$.
  Does greater market depth imply that prices reveal less information about fundamental value?
- **Price informativeness:** Dealers observe y, and $P(y) = \mathbb{E}[V \mid y]$. By joint normality, the remaining uncertainty about V is:

$$\mathrm{Var}(V \mid y) = \underbrace{\mathrm{Var}(V)}_{\text{prior uncertainty}} - \underbrace{\mathrm{Cov}(V,y)^2/\mathrm{Var}(y)}_{\text{uncertainty resolved by order flow}} = \sigma_V^2 - \beta^2\sigma_V^4/2\sigma_u^2 = \sigma_V^2/2.$$

  ↪ Observing order flow resolves **half of the prior variance**, regardless of σ_u.

! This does not mean that half of each realized order is informed. Why does informativeness remain unchanged?
  - More noise trading ($\sigma_u \uparrow$) makes an order less informative *for a given insider strategy*.
  - However, the insider responds with $\beta \uparrow$, keeping $\mathrm{Var}(x) = \mathrm{Var}(u) = \sigma_u^2$.

  ↪ Relative signal and noise variances remain unchanged.

- **Insider's expected profit:** Since $x = \beta(V - \mu)$ and $\mathbb{E}[ux] = 0$:

$$\mathbb{E}[(V - P(y))x] = \mathbb{E}[(V-\mu)x] - \lambda\mathbb{E}[x^2] + 0 = \beta(1-\lambda\beta)\sigma_V^2 = \boxed{\frac{\sigma_u\sigma_V}{2}}, \qquad \text{as } \lambda\beta = \frac{1}{2}.$$

  ↪ More noise trading increases insider profits by allowing larger trades with lower price impact.

▷ **Market depth ≠ price informativeness:** Noisier trading reduces price impact, but induces stronger informed trading. These effects offset in the information revealed by prices.

Inventory Risk, Spreads, and Market Depth

- In Glosten-Milgrom/Kyle, dealers are risk-neutral and face no inventory costs. *What if dealers must hold risky positions?*
- **Inventory risk:** Dealers absorb imbalances between buyers and sellers.

  A dealer holding a large inventory faces losses if the asset price falls before the position can be unwound.

  ↪ A risk-averse dealer with a **long** (short) inventory may lower (increase) both quotes: a lower ask encourages customers to buy from the dealer, while a lower bid discourages additional customer sales.

- **Two distinct mechanisms:**
  - **Adverse selection:** Orders reveal information about fundamental value.
  - **Inventory risk:** Dealers require compensation for carrying risky positions (i.e., quotes are risk-adjusted to encourage trades that reduce unwanted exposure).

Dimension of market liquidity	Economic interpretation
Tightness (spread)	Cost of immediately buying and reselling a small position.
Depth (price impact)	How much the execution price changes with the size of an order.
Resiliency	How quickly prices and available liquidity recover after an order imbalance.

↪ A narrow bid-ask spread does not guarantee low trading costs: large orders may still move prices substantially in shallow mkt.

⊵ **Liquidity is multidimensional:** Trading costs depend on both the price quoted and the mkt's ability to absorb/recover.

Trading Costs: Amihud–Mendelson (1/4)

- *How does the cost of trading (rather than information) affect the value investors assign to an asset?*
- Consider an asset paying a constant dividend flow $d > 0$ per year, with continuous discount rate $\rho > 0$.
  - The investor buys at the stationary ask P and sells at the bid $(1 - s)P$, with $s \in [0, 1)$.
  - Liquidation occurs at a random time $T \sim \text{Exp}(h)$, so $\mathbb{E}[T] = 1/h$.
- **Value of holding the asset:** 1) dividends are received until liquidation at T, 2) followed by resale at the bid:

$$\mathbb{E}\left[\underbrace{\int_0^T e^{-\rho t} d \, \mathrm{d}t}_{\text{1) dividends until liquid.}} + \underbrace{e^{-\rho T}(1-s)P}_{\text{2) resale proceeds at } T}\right].$$

- Since $T \sim \text{Exp}(h)$: $\Pr(T > t) = e^{-ht}$, $\quad f_T(t) = he^{-ht}$.
  - ↪ $\Pr(T > t)$ is the probability the investor is still holding the asset at date t.
  - ↪ $f_T(t)$ is the density of liquidating exactly at date t.
- **Expected discounted dividends:** dividend at t is received only if the investor still holds the asset:

$$\mathbb{E}\left[\int_0^T e^{-\rho t} d \, \mathrm{d}t\right] = \int_0^\infty e^{-\rho t} d \, \Pr(T > t) \, \mathrm{d}t = d \int_0^\infty e^{-(\rho+h)t} \mathrm{d}t = \boxed{\frac{d}{\rho + h}}.$$

  - ↪ Recall: for $a > 0$, $\displaystyle\int_0^\infty e^{-at} \mathrm{d}t = \left[-\frac{e^{-at}}{a}\right]_0^\infty = \frac{1}{a}$.

Trading Costs: Amihud–Mendelson (2/4)

- **Expected discounted resale:** The investor sells at the random date T. Averaging over possible liquidation dates:

$$\mathbb{E}[e^{-\rho T}(1-s)P] = (1-s)P\int_0^\infty e^{-\rho t} f_T(t)\,\mathrm{d}t = (1-s)P\int_0^\infty e^{-\rho t} h e^{-ht}\,\mathrm{d}t = (1-s)Ph/(\rho+h).$$

- Hence, the expected discounted proceeds from holding the asset are: $(d + h(1-s)P)/(\rho+h)$.
- In a stationary one-clientele reservation-value benchmark, the purchase price equals the expected discounted proceeds:

$$P = \frac{d + h(1-s)P}{\rho+h} \Longrightarrow P(\rho+h) = d + hP - hsP \Longrightarrow \boxed{P = \frac{d}{\rho+hs}}.$$

- **Dividend yield** (dividend over price):

$$d/P = \underbrace{\rho}_{\text{discount rate}} + \underbrace{hs}_{\text{annualized trading-cost burden}}.$$

  ρ: discounting of future cash flows; h: liquidation intensity; s: proportional resale loss.
- **Comparative statics:** Holding d and ρ fixed:

$$\frac{\partial P}{\partial s} = -\frac{dh}{(\rho+hs)^2} < 0, \qquad \frac{\partial P}{\partial h} = -\frac{ds}{(\rho+hs)^2} < 0 \quad (s>0).$$

  ↪ $s\uparrow$: A larger spread lowers the resale value, hence the purchase price.
  ↪ $h\uparrow$: Shorter expected holding periods (higher liquidation intensity) make the same spread more costly.

▷ The same asset is more valuable to long-horizon investors, because they expect to incur the resale cost less frequently.

Who Holds Illiquid Assets? (1/2)

- Investor j with liquidation intensities $h_j > 0$. Assets i with dividend yields $y_i = d_i/P_i$ and spreads s_i.
  - Type j chooses the asset offering the highest yield net of expected trading costs: $r_j^* = \max_i\{y_i - h_j s_i\}$.
  - A larger h_j means a shorter expected holding period and therefore greater sensitivity to the spread.
- **When does type j hold asset i?** Only when asset i is as attractive as the best available alternative:

$$y_i - h_j s_i = r_j^* \qquad \Longleftrightarrow \qquad y_i = r_j^* + h_j s_i.$$

- By definition of r_j^*, every asset must satisfy:

$$y_i - h_j s_i \leq r_j^* \qquad \Longleftrightarrow \qquad y_i \leq r_j^* + h_j s_i, \qquad \forall j.$$

- Since asset i has positive supply, some investor must actually hold it and attain equality:

$$\boxed{y(s_i) = \min_j\{r_j^* + h_j s_i\}.}$$

↪ At a given spread, different investor types imply different required dividend yields. The market yield is determined by the investor willing to hold the asset at the **lowest** yield; all other investors prefer their alternatives.

≥ Illiquid assets are held by the investors for whom trading costs matter the least (i.e., the least sensitive).

Who Holds Illiquid Assets? (2/2)

- **Why do investors sort by holding horizon?** Suppose type j holds asset a and type k holds asset b, with $h_j > h_k$. Since each type prefers the asset it holds:

$$y_a - h_j s_a \geq y_b - h_j s_b, \qquad y_b - h_k s_b \geq y_a - h_k s_a.$$

- Adding the two inequalities:

$$(y_a - y_b) - h_j(s_a - s_b) \geq 0, \quad (y_b - y_a) - h_k(s_b - s_a) \geq 0 \Longrightarrow \quad \boxed{(h_j - h_k)(s_b - s_a) \geq 0}.$$

- Since $h_j > h_k$, it follows that: $s_b \geq s_a$.
  - ↪ The investor k with the **longer horizon** (smaller h) holds the asset b with the **higher spread**.
  - ↪ The investor j with the **shorter horizon** avoids high-spread assets, because the trading cost is incurred more frequently.
- **Liquidity premia:** higher spreads require higher dividend yields.
  However, as spreads rise, the marginal holders are investors with longer horizons, who are less sensitive to additional trading costs.
  ↪ The yield-spread relation remains upward sloping, but it becomes **flatter**
  as the market reallocates illiquid assets toward long-horizon investors.

Acharya-Pedersen (1/4): After Trading Costs

- Previously, investors required compensation for expected trading costs, rather than unexpected:
  what if selling costs are uncertain and become particularly high in bad market states?
- An investor at t buys one share at $P_{i,t}$, receives dividend $D_{i,t+1}$, and must sell at $t+1$.
  - Selling incurs a **random cost** $C_{i,t+1}$ per share: the investor receives $P_{i,t+1} - C_{i,t+1}$ rather than $P_{i,t+1}$.

  ↪ The total payoff after liquidation costs is $P_{i,t+1} + D_{i,t+1} - C_{i,t+1}$.
- Divide each component by the purchase price:

$$R^g_{i,t+1} = \underbrace{\frac{P_{i,t+1} + D_{i,t+1}}{P_{i,t}}}_{\text{gross return before costs}}, \qquad c_{i,t+1} = \underbrace{\frac{C_{i,t+1}}{P_{i,t}}}_{\text{cost per dollar invested}}, \qquad R^{g,n}_{i,t+1} = R^g_{i,t+1} - c_{i,t+1}.$$

- With $r_{i,t+1} = R^g_{i,t+1} - 1$, net simple returns: $r^n_{i,t+1} = r_{i,t+1} - c_{i,t+1}$, $\mathbb{E}_t[r^n_{i,t+1}] = \mathbb{E}_t[r_{i,t+1}] - \mathbb{E}_t[c_{i,t+1}]$.

  ↪ Subtracting 1 removes the principal vs. subtracting c_i accounts for the liquidation cost.
- **Expected costs versus liquidity risk:**
  - **Expected cost:** $\mathbb{E}_t[c_i]$ lowers the return investors expect to receive, even when the cost is predictable.
  - **Liquidity risk:** Unexpected increases in c_i reduce net returns.

    ↪ especially costly if they occur when the market performs poorly.

▷ Investors care about the **return after selling costs**, not merely the asset's quoted return.
Both expected costs and their co-movement with market conditions may affect prices.

AP (2/4): Portfolio Choice and Market Clearing

- *How do investors' preferences for net returns determine equilibrium expected returns?*
- **Investor's problem:** (Suppress t for simplicity). Let $r^n = r - c$ denote the vector of net simple returns, with:

$$e = \mathbb{E}[r^n] - r_f \mathbf{1}, \qquad \Sigma = \mathrm{Var}(r^n) \quad \text{(nonsingular)}.$$

  Under CARA preferences and normally distributed net returns, investor j with absolute risk aversion $a_j > 0$ chooses **dollar positions** $x^j \in \mathbb{R}^N$:

$$\max_{x^j} \quad (x^j)'e - \frac{a_j}{2}(x^j)'\Sigma x^j, \qquad \text{FOC: } e - a_j \Sigma x^j = 0 \quad \Longrightarrow \quad \boxed{x^j = \frac{1}{a_j}\Sigma^{-1} e}.$$

  ↪ Expected net excess returns encourage investment; covariance risk raises the marginal cost of holding risky positions.
  ↪ Investors can borrow/lend at the safe rate. No short sales.

- **Market clearing:** Aggregate risky dollar holdings must equal the market capitalization vector:

$$m = \sum_j x^j, \qquad V_M = \mathbf{1}'m > 0, \qquad w_M = m/V_M.$$

  V_M is total risky market capitalization, w_M contains the market portfolio's value weights, $\mathbf{1}'w_M = 1$.

- Define aggregate risk aversion $a = (\sum_j 1/a_j)^{-1}$:

$$m = \sum_j \frac{1}{a_j}\Sigma^{-1} e = \frac{1}{a}\Sigma^{-1} e \quad \Longrightarrow \quad \boxed{e = a\Sigma m = aV_M \Sigma w_M}.$$

▷ **Equilibrium risk pricing:** Market clearing links expected net excess returns (e) to covariance with the aggregate risky portfolio, rather than with any individual investor's holdings.

AP (3/4): The Net-Return CAPM

- Using the market value weights w_M:

$$r_M = w_M' r, \qquad c_M = w_M' c, \qquad \boxed{r_M^n = r_M - c_M = w_M' r^n}.$$

  $\hookrightarrow$ The market return must also be measured net of its aggregate liquidation costs.

- From $e = aV_M \Sigma w_M$, the expected net excess return on asset i satisfies:

$$e_i = aV_M (\Sigma w_M)_i = aV_M \underbrace{\mathrm{Cov}(r_i^n, r_M^n)}_{\text{covariance w net mkt return}}.$$

- Multiplying the same equilibrium condition by w_M':

$$e_M := w_M' e = aV_M w_M' \Sigma w_M = aV_M \underbrace{\mathrm{Var}(r_M^n)}_{\text{net mkt-return variance}}.$$

- **To recover the CAPM**, substitute aV_M using the latter into the former:

$$\boxed{\mathbb{E}_t[r_i - c_i] - r_f = \underbrace{\left(\mathbb{E}_t[r_M - c_M] - r_f\right)}_{\mathrm{RP}^n_{M,t}} \underbrace{\frac{\mathrm{Cov}_t(r_i - c_i, r_M - c_M)}{\mathrm{Var}_t(r_M - c_M)}}_{\text{net-return beta}}}.$$

$\hookrightarrow$ Assets with greater exposure to adverse movements in **net market returns** require higher expected net returns.

$\trianglerighteq$ Unlike classic CAPM, the covariance includes fluctuations in both asset returns and liquidation costs.

AP (4/4): Four Covariance Terms

- **Decomposing net-return risk:**

$$\mathrm{Cov}_t(r_i - c_i, r_M - c_M) = \mathrm{Cov}_t(r_i, r_M) + \mathrm{Cov}_t(c_i, c_M) - \mathrm{Cov}_t(r_i, c_M) - \mathrm{Cov}_t(c_i, r_M).$$

- Since $\mathbb{E}_t[r_i - c_i] = \mathbb{E}_t[r_i] - \mathbb{E}_t[c_i]$, define $D_t = \mathrm{Var}_t(r_M - c_M) > 0$. Then:

$$\boxed{\mathbb{E}_t[r_i] - r_f = \mathbb{E}_t[c_i] + \frac{\mathrm{RP}^n_{M,t}}{D_t}\Big[\mathrm{Cov}_t(r_i, r_M) + \mathrm{Cov}_t(c_i, c_M) - \mathrm{Cov}_t(r_i, c_M) - \mathrm{Cov}_t(c_i, r_M)\Big].}$$

  - $+\mathrm{Cov}(r_i, r_M)$ - **Mkt risk:** If the asset performs poorly when the mkt falls, its return exposure raises the premium.
  - $+\mathrm{Cov}(c_i, c_M)$ - **Common liquidity risk:** If the asset becomes expensive to sell when the entire mkt is illiquid, investors require additional compensation.
  - $-\mathrm{Cov}(r_i, c_M)$ - **Return sensitivity to mkt liquidity:** An asset paying more when market selling costs rise provides insurance. A positive covariance reduces its required premium.
  - $-\mathrm{Cov}(c_i, r_M)$ - **Liquidity sensitivity to market returns:** If the asset's selling costs rise when market returns fall, this covariance is negative. Subtracting it raises the premium.

! The covariance terms need not all have the same sign. Their contribution to the premium depends on how returns and costs co-move. In this equilibrium, $\mathrm{RP}^n_{M,t}/D_t > 0$.

▷ **Expected illiquidity versus liquidity risk:** $\mathbb{E}_t[c_i]$ compensates for the expected cost of selling; the three additional liquidity-risk exposures compensate for unfavorable co-movement with aggregate market conditions.

Transaction Costs and the No-Trade Region

- Recall Session 3: investors could continuously rebalance their portfolios.

  Magill-Constantinides: Let $J(b, n, S, t)$ denote continuation utility, with cash b, shares n, and mid-price S.
  - Buying one share costs $(1+\tau)S$ vs. selling one share yields $(1-\tau)S$ (with $\tau > 0$).
  - J_b is the marginal value of cash; J_n is the marginal value of an additional share.

- By Taylor expansion, for small changes $(\Delta b, \Delta n)$, holding (S, t) fixed:

$$J(b+\Delta b, n+\Delta n, S, t) - J(b, n, S, t) \simeq \underbrace{J_b \Delta b}_{\text{effect of changing cash}} + \underbrace{J_n \Delta n}_{\text{effect of changing shares}}.$$

  - **Buy?** Buying $\varepsilon > 0$ shares implies $\Delta b = -(1+\tau)S\varepsilon$ and $\Delta n = \varepsilon$. Substituting into the expansion:

$$\Delta J^B \simeq J_b[-(1+\tau)S\varepsilon] + J_n\varepsilon = \varepsilon \underbrace{[J_n - (1+\tau)SJ_b]}_{\text{share benefit minus cash cost}} \implies \text{Buy if } J_n > (1+\tau)SJ_b$$

  - **Sell?** Selling $\varepsilon > 0$ shares implies $\Delta b = (1-\tau)S\varepsilon$ and $\Delta n = -\varepsilon$:

$$\Delta J^S \simeq J_b[(1-\tau)S\varepsilon] - J_n\varepsilon = \varepsilon \underbrace{[(1-\tau)SJ_b - J_n]}_{\text{cash received minus share value}} \implies \text{Sell if } J_n < (1-\tau)SJ_b$$

- **When nothing is optimal?** Neither should increase utility: $J_n - (1+\tau)SJ_b \leq 0$, $(1-\tau)SJ_b - J_n \leq 0$.
  For $J_b > 0$, combining these: $\underbrace{(1-\tau)S}_{\text{bid}} \leq \underbrace{J_n/J_b}_{\text{marginal share value in cash}} \leq \underbrace{(1+\tau)S}_{\text{ask}}$.

▷ **No-trade region:** The investor does not rebalance while the marginal value of a share lies between the bid and ask.

TBD